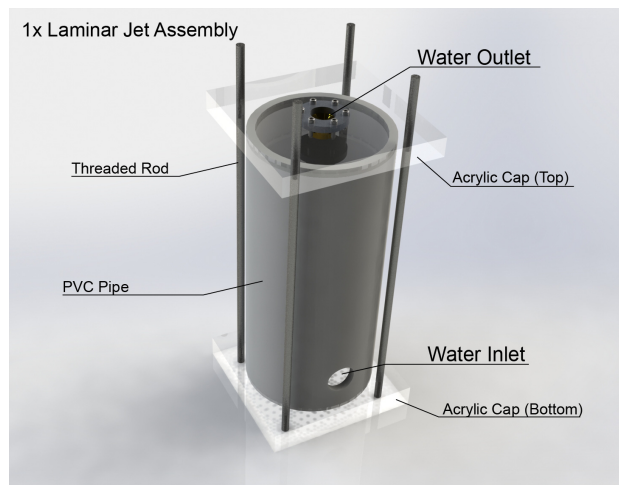
ChE 2O04 Tutorial #6

Tutorial Problem #1

As was mentioned in the lectures, a laminar water jet creates 'glass-like' rods of water that are used in public displays (e.g. at the Detroit airport) and in backyard landscaping. There are numerous websites and Youtube videos showing you how to make one of these devices from common materials such as a PVC pipe filled with drinking straws - for example: https://alanm52.wordpress.com/2014/02/28/laminar-flow-water-jet/



Imagine that you are given some instructions based on a 100 mm (inside diameter) PVC pipe that is packed with drinking straws (6 mm outside diameter).

a) Prove that for water at 10 °C flowing through the device at 90 L/min that the flow is indeed laminar. You can assume that no flow goes in the gaps between the straws. Also, use this website to calculate the maximum number of drinking straws that can fit into the PVC pipe: https://www.engineeringtoolbox.com/smaller-circles-in-larger-circle-d_1849.html

b) Prove that in the absence of the drinking straws that the flow becomes turbulent (and thus is no longer a laminar water jet).

Tutorial Problem #2

On slide 20 of the PDF_06 notes, we gave you the following equation for terminal velocity of a spherical particle:

$$v_t = \sqrt{\frac{4}{3}\frac{\left(\rho_{particle} - \rho_{fluid}\right) g D_p}{C_D \rho_{fluid}}}$$

a) Prove that the equation is dimensionally consistent.
b) Calculate the terminal velocity (m/s) of a grain of sand (diameter = 1 mm, density = 2300 kg/m^3) in ambient temperature air (i.e. 20 °C).